

F78PB Probability and Statistics B

Statistical models and Estimators

Construction and estimation of statistical models

A mathematical model is a simplified representation of a real phenomenon through a *mathematically expressed* system.

This can be a statistical model when any uncertainty in the system is expressed through *probabilities* and *statistical distributions*.

The process through which we estimate unknown quantities (*parameters*) in statistical models using available information (*data*) is called statistical inference.

Example

Suppose that a farmer has to choose between two fertilizers (A and B). Before placing a large order for either he obtains samples of each and apply A to 5 'test' plots and B to 5 plots of a certain crop. Then at the end of the season he measures the yield of the crop in *kgs* in each plot. The results are:

Fertiliser A: 12.5, 14.4, 15.1, 11.9, 13.1

Fertiliser B: 12.6, 13.2, 10.4, 10.9, 12.8
For fertilisers A and B the average yields are 13.4 and 11.98, respectively.
Does this tell us that A is better than B and he should place a large order for A?

Example

A farmer has to choose between two fertilizers (A and B). He obtains samples of each and apply A to 5 'test' plots and B to 5 plots of a certain crop. Then he measures the yield of the crop in *kgs* in each plot. The results are:

$$\underline{x}_A = (12.5, 14.4, 15.1, 11.9, 13.1), \qquad \bar{x}_A = 13.4,$$
$$\underline{x}_B = (12.6, 13.2, 10.4, 10.9, 12.8), \qquad \bar{x}_B = 11.98.$$

Some points to consider:

- Data indicate that 'yield' is something that varies from plot to plot even if the same fertilizer is used.
- If we repeated the experiment we would surely obtain different measurements and, plausibly, could come to a different conclusion regarding which was better, i.e. it is possible that B may be better!

Remarks:

- Statistical inference is the process whereby we will *quantify* how strong the evidence is that A is really better than B.
  We can then decide whether to order A, or whether, perhaps, we should do further experiments before making a decision.
- Not difficult to think of all kinds of scenarios that are analogous to this example: testing effectiveness of different drugs, comparing lifetimes of different brands of a component.
  - Every day we have to make potentially important decisions in the face of limited evidence;
  - statistics gives us a set of tools for trying to do this.

The Concept of Probability

What is the role of Probability Theory?

- Provides a framework to model how observations arise.
- Quantifies uncertainty in conclusions.

Illustrative Question: Can we assign a probability to the statement: *"A is better than B"?*

Fertilizer Example (continued):

- Measurements for fertilizer B follow:

$$X \sim \mathrm{N}(\mu_B, 1), \quad \mu_B \text{ unknown.}$$

- Observed sample mean: 11.98, sample size: $n = 5$.

Observed 95% **Confidence Interval for** μ_B**:**

$$\left(11.98 - 1.96 \cdot \frac{1}{\sqrt{5}},\ 11.98 + 1.96 \cdot \frac{1}{\sqrt{5}}\right) \approx (11.1,\ 12.8).$$

Interpretation: This interval expresses the degree of certainty about plausible values of μ_B, using probability.

Interpretations of probability

In the previous example the '95%' is indeed a probability. But of what event?

To a pure *frequentist*, the only meaningful definition of probability of an event is the frequency with which it occurs over a *long sequence of independent trials*.

Thus statements such as:

- the probability that a '6' is scored when a fair die is rolled is $\frac{1}{6}$
- the probability that a 'H' is uppermost when a fair coin is tossed is $\frac{1}{2}$

would be perfectly meaningful to a frequentist because there is a clearly defined, repeatable experiment in each case.

The probability has an interpretation as a frequency.

For example

- the ‘probability’ that I voted Labour in the last election
- the ‘probability’ that Celtic will win the Scottish League this year

would *not* be recognised by a frequentist as true probabilities.

These deal with propositions that must either be *true or false*.

There is no sequence of *identical* repeatable experiments for which these probabilities are frequencies.

For the last example, there is only one Scottish Premier League in the current season. Even though the SPL is played every year, the teams and their personnel change from year to year.

However,

- these probabilities - which express the degree of belief in a proposition regarding the world and are known as *subjective probabilities* - *are* meaningful;
- for example, a bookmaker, when setting the odds of Celtic winning the SPL this year, must consider his subjective probability that they will win.

Fertilizer Example continued

Recall: In the fertilizer example, we calculated a 95% CI for μ_B to be:

$$(11.1,\ 12.8).$$

Interpretation of 95%:

- The 95% must correspond to the frequency of some event over repeated independent trials. Which event? Which experiment?

Repeatable experiment:

- Sow 5 plots, fertilize with B, measure yields.
- Assume observations $Y_1, \ldots, Y_5 \sim \mathrm{N}(\mu_B, 1)$, with μ_B unknown but constant across trials.

What event occurs with frequency 0.95? Is it $\mu_B \in (11.1, 12.8)$? **No!** Since μ_B never changes, this event has frequency 0 or 1.

Correct event: The confidence interval we calculate contains μ_B.

Classical interpretation: Since our experiment is not atypical, we can be 95% confident that CI $(11.1, 12.8)$ contains μ_B.

The main elements of frequentist (classical) inference

The experiment

This is a procedure that can be repeated independently many times where the outcome of the experiment is subject to uncertainty.

The outcome of the experiment can be considered to be a random variable X, i.e. a measurement on some real-valued quantity.

Typical examples of an experiment are:

1. Select an individual randomly (with replacement) from a large population and determine whether they carry a certain gene.
2. Select a component randomly from a large population and measure the lifetime.
3. Observe the night sky and measure the time elapsed before the first meteor appears.

The statistical model

The statistical model is simply the *distribution* that we assume for the observation X.

Exercise: Suggest suitable statistical models for the experiment examples on the previous slide.

The statistical model is also referred to as the *population distribution* of the measurement X.

- Repeating the 'basic' experiment several times we obtain a sequence of measurements $X_1, X_2, \ldots, X_n$.
- When we have *independence* of the measurements we refer to them as a *random sample of size n*. This is equivalent to $X_1, X_2, \ldots, X_n$ being i.i.d. with probability function f_X.
- With independence, the multivariate p.d.f. or p.m.f. of $\underline{X} = (X_1, X_2, \ldots, X_n)$ is

$$f_{\underline{X}}(x_1, x_2, \ldots, x_n) = f_X(x_1) \times f_X(x_2) \times \ldots \times f_X(x_n)$$

Model parameters

- Usually, although we might be sure of the family from which the distribution of X comes, we don't know the distribution's parameters.
- The main point of doing the experiment is (usually) to find out more about the parameters or some other fixed characteristic of the population distribution such as its *mean* or *variance*.
- We call this process *estimation*.
  In the fertilizer example we want to estimate $\mu_A - \mu_B$ to help us choose between the 2 fertilizers.

Exercise: make sure you know the mean and variance of the Gamma, Exponential, Normal, Binomial, and Poisson distributions in terms of their parameters.

Estimation

We carry out an experiment that yields a random sample $X_1, \ldots, X_n$ from some population distribution.
We wish to use the data to estimate a quantity θ which could be a population moment (*e.g.* mean or variance) or a parameter.

Since our estimate depends on the data, it must be some *function of the random sample* and we can write it as

$$\hat{\theta}(X_1, \ldots, X_n)$$

We refer to $\hat{\theta}$ as an estimator of θ.

Sometimes there will be a very natural choice for the function $\hat{\theta}$.

Distinguishing Estimator Notation

- The estimator

$$\hat{\theta}(\underline{X})$$

  is a **random variable** because it is a function of the random sample

$$\underline{X} = (X_1, X_2, \ldots, X_n.)$$

- Its value changes from one experiment (or sample) to another.
- Make sure you distinguish:

$$\hat{\theta}(\underline{X}) \quad \text{(random variable)}$$

$$\hat{\theta}(\underline{x}) \quad \text{(observed value)}$$

Sample mean and sample variance

Suppose we wish to estimate the population mean (usually called μ) from a random sample $X_1, \ldots, X_n$. Then a natural estimator is the *sample mean*

$$\hat{\mu}(\underline{X}) = \bar{X} = \frac{\sum X_i}{n}.$$

If we need to estimate the population variance σ^2 from the random sample, then a natural estimator to use here would be the *sample variance*:

$$\widehat{\sigma}^2(\underline{X}) = S^2 = \frac{1}{n-1}\left\{\sum X_i^2 - \frac{(\sum X_i)^2}{n}\right\}$$

Why are $\bar{X}$ and S^2 sensible choices as estimators for μ and σ^2?

Properties of estimators

We conduct an experiment which yields a random sample $X_1, \ldots, X_n$ from a population. We wish to estimate some population quantity θ using an estimator $\hat{\theta}(\underline{X})$.

How can we decide whether this choice of estimator is sensible?

Bias of an estimator

We refer the distribution of $\hat{\theta}(\underline{X})$ as its sampling distribution.
It tells us how $\hat{\theta}(\underline{X})$ will vary over a large number of independent experiments.

Now if $\hat{\theta}(\underline{X})$ is a sensible estimator for θ then

- the values it typically takes should be close to θ in some sense
- if we conduct a number of identical experiments, each giving an estimate $\hat{\theta}(\underline{X})$, then the values should be scattered around the true value of θ.

This leads naturally the concept of bias.
We define the bias of an estimator (for θ) $\hat{\theta}(\underline{X})$ to be:

$$\text{Bias}(\hat{\theta}(\underline{X})) = \mathbb{E}\left[\hat{\theta}(\underline{X}) - \theta\right].$$

If $\text{Bias}(\hat{\theta}(\underline{X})) = 0$ then this means that the expectation of $\hat{\theta}(\underline{X})$ is precisely θ and we say that $\hat{\theta}(\underline{X})$ is unbiased.

Example

Let us return to the estimation of the population mean μ from random sample $X_1, ..., X_n$. What is the bias of the estimator $\hat{\mu}(\underline{X}) = \bar{X}$?
Also, show that the sample variance, S^2 is an unbiased estimator of the population variance σ^2.
Solution: ...

Note that this is true so long as the population mean and variance exist - it doesn't matter what the precise form of the distribution F_X is.

Recall from the *central limit theorem* (CLT):
The sample mean $\bar{X}$ has *approximately* the $\mathrm{N}\left(\mu, \frac{\sigma^2}{n}\right)$ distribution for large n.

- This is exact if $X_i \sim N(\mu, \sigma^2)$.
- Otherwise the approximation get better as $n \to \infty$.

Example

Suppose now that the observations $X_1, ..., X_n$ are a random sample from a Gamma (α, β) distribution. (This could arise if our observations were lifetimes of components or survival times of individuals.) Identify the precise form of the distribution of $\bar{X}$ in this case.
Solution: ...

Not all unbiased estimators are good

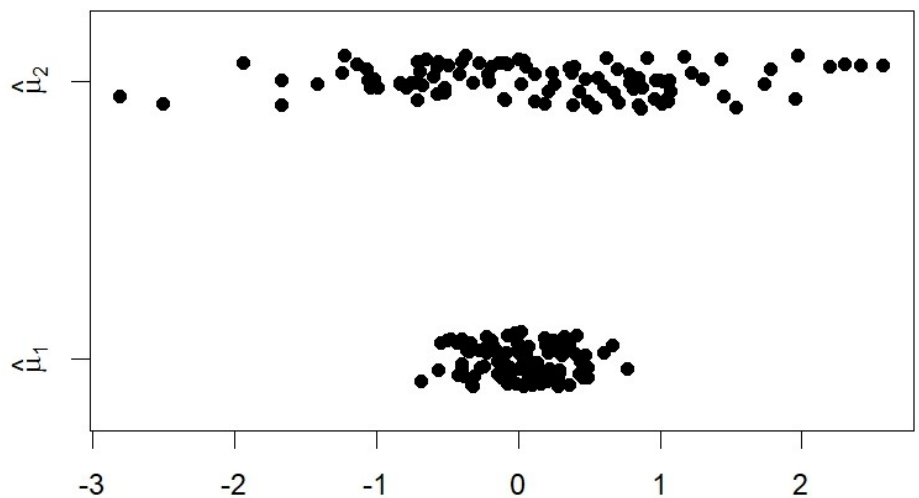
Consider estimating the population mean of a random variable $X \sim \mathrm{N}(\mu, 1)$. Suppose we have a random sample $X_1, \ldots, X_n \overset{i.i.d.}{\sim} \mathrm{N}(\mu, 1)$. Consider the following estimators:

$$\hat{\mu}_1(\underline{X}) = \bar{X}, \quad \hat{\mu}_2(\underline{X}) = X_1.$$

We can calculate the distribution of these estimators:

$$\hat{\mu}_1(\underline{X}) = \bar{X} \sim \mathrm{N}\left(\mu, \frac{1}{n}\right), \quad \hat{\mu}_2(\underline{X}) = X_1 \sim \mathrm{N}(\mu, 1).$$

Both estimators are unbiased. Which estimator is more useful?



Mean square error

For an estimate $\hat{\theta}(\underline{X})$ of θ we want the random error $\hat{\theta}(\underline{X}) - \theta$ to be small.

We define the *mean square error (MSE)* of $\hat{\theta}(\underline{X})$ as

$$\text{MSE}(\hat{\theta}(\underline{X})) = \mathbb{E}[(\hat{\theta}(\underline{X}) - \theta)^2].$$

This can be written as

$$\text{MSE}(\hat{\theta}(\underline{X})) = \text{Bias}(\hat{\theta}(\underline{X}))^2 + \mathbb{V}\text{ar}(\hat{\theta}(\underline{X})) \quad [\textit{Show}]$$

For unbiased estimators: $\text{MSE}(\hat{\theta}(\underline{X})) = \mathbb{V}\text{ar}(\hat{\theta}(\underline{X}))$.

Example

Find the MSE of the sample mean $\bar{X}$ as an estimator of the population mean μ.

Solution:
We have seen that $\bar{X}$ is an unbiased estimator of μ.
It follows from the equation above that $\text{MSE}(\bar{X}) = \mathbb{V}\text{ar}(\bar{X}) = \frac{\sigma^2}{n}$.
(You should be able to prove this!!)

Efficiency

We can use the MSE as a measure to compare different estimators. If $\hat{\theta}_1(\underline{X})$ and $\hat{\theta}_2(\underline{X})$ are estimators of θ, the *(relative) efficiency* of $\hat{\theta}_2(\underline{X})$ compared to $\hat{\theta}_1(\underline{X})$ is given by the ratio

$$\frac{\mathrm{MSE}(\hat{\theta}_1(\underline{X}))}{\mathrm{MSE}(\hat{\theta}_2(\underline{X}))}.$$

We say that $\hat{\theta}_1$ is more efficient than $\hat{\theta}_2$ if $\mathrm{MSE}(\hat{\theta}_1(\underline{X})) < \mathrm{MSE}(\hat{\theta}_2(\underline{X}))$.

Example

Consider a random sample (X_1, X_2) from a population with mean μ and variance σ^2. Now consider the following as estimators of μ:
$\hat{\mu}_1(X_1, X_2) = \bar{X}$ and $\hat{\mu}_2(X_1, X_2) = \frac{X_1}{3} + \frac{2X_2}{3}$.
Determine which of the two estimators is more efficient.

Solution:
Both of these are unbiased ($\mathbb{E}(\hat{\mu}_1(X_1, X_2)) = \mathbb{E}(\hat{\mu}_2(X_1, X_2)) = \mu$).
It follows that $\text{MSE}(\hat{\mu}_1(X_1, X_2)) = \mathbb{V}\text{ar}(\hat{\mu}_1(X_1, X_2)) = \frac{\sigma^2}{2}$.

Also,
$\text{MSE}(\hat{\mu}_2(X_1, X_2)) = \mathbb{V}\text{ar}(\hat{\mu}_2(X_1, X_2)) = \frac{\sigma^2}{9} + \frac{4\sigma^2}{9} = \frac{5\sigma^2}{9} > \text{MSE}(\hat{\mu}_1(X_1, X_2))$.

Therefore $\hat{\mu}_1$ is more efficient that $\hat{\mu}_2$.

Example

In an experiment a bacteriologist must measure the temperature, θ of a growth chamber. She has two thermometers of differing quality. The first returns a random quantity $T_1 = \theta + E_1$ where E_1 is a random error with mean zero and variance σ_1^2. The second returns $T_2 = \theta + E_2$ where E_2 has zero mean and variance σ_2^2. Suppose further that E_1 and E_2 are independent. She decides to combine the two measurements via the estimator:

$$\hat{\theta}(T_1, T_2) = aT_1 + (1 - a)T_2$$

where $a \geq 0$. How should a be chosen to give the most efficient estimator of this form?

Solution: ...

Standard Error

As already mentioned, good estimators should have small variance, i.e., *large precision*.

The square root of the variance of an estimator $\hat{\theta}(\underline{X})$ is called its standard error:

$$\text{s.e.}(\hat{\theta}(\underline{X})) = \{\mathbb{V}\text{ar}(\hat{\theta}(\underline{X}))\}^{1/2}.$$

It is especially useful when comparing estimators.

Example

On the day of HW's Students' Association elections, one representative wants to predict the proportion of students who voted for candidate A (before results are known!).

She selects a sample of m students *at random*, who have just voted, and asks them anonymously whether they voted for candidate A.

She considers the outcome as a random sample $\underline{X} = (X_1, X_2, \ldots, X_m)$ from a Bernoulli(p) distribution, where:

$$X_i = \begin{cases} 1 & \text{if } i^{th} \text{ student voted for A,} \\ 0 & \text{otherwise.} \end{cases}$$

Question: Is there an obvious estimator for p?

Yes, let

$$\hat{p}(\underline{X}) = \frac{\sum X_i}{m}$$

(*i.e.* the number of students in the sample who voted for A, over the total number of voters in the sample).

Find the bias, standard error and MSE of $\hat{p}(\underline{X})$.

Solution: ...